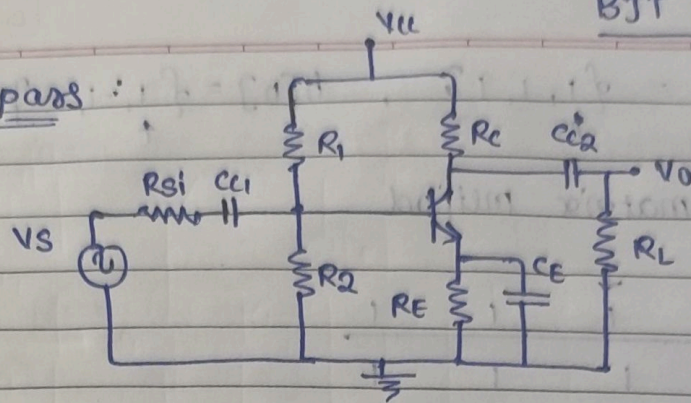


$$\left[g_m = \frac{I_c}{V_T} \quad r_o = \frac{V_A}{I_c} \quad r_{\pi} = \frac{\beta V_T}{I_c} \right]$$

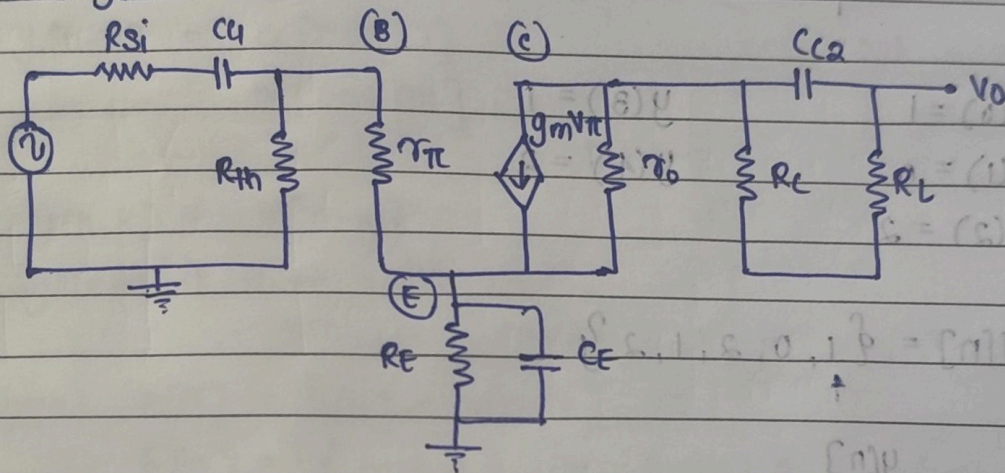
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BJT Amplifiers

→ CE - Bypass



• LF Analysis:



$$a) f_{u1} = \frac{1}{2\pi \cdot R_{eq} \cdot C_{c1}}$$

$$= \frac{1}{2\pi [R_{si} + (R_{th} \parallel r_{\pi})] C_{c1}}$$

here,

$$R_{eq} = R_{si} + (R_{th} \parallel r_{\pi})$$

$$R_{eq} = R_{si} + (R_{th} \parallel [r_{\pi} + (1+\beta)R_E])$$

BYPASS

UNBYPASS.

$$b) f_{u2} = \frac{1}{2\pi (R_C + R_L) C_{c2}}$$

$$c) f_{uE} = \frac{1}{2\pi [R_E \parallel \frac{(R_{si} \parallel R_{th} + r_{\pi})}{1+\beta}] C_E}$$

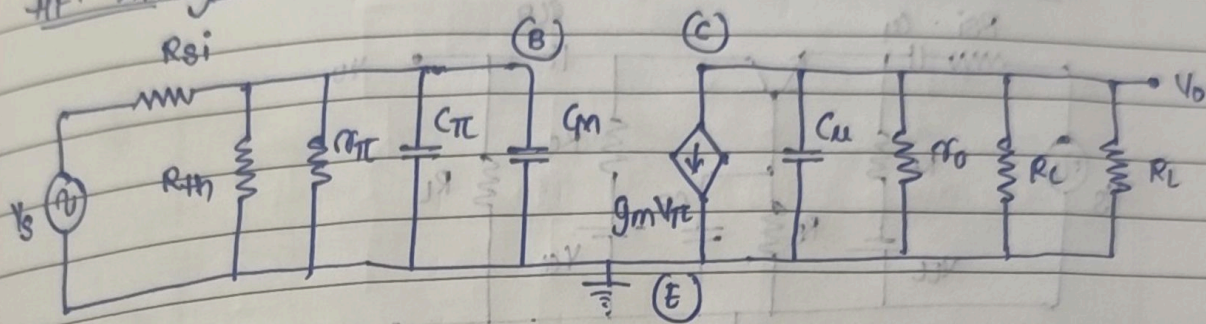
$$C_{bc} \rightarrow C_u$$

$$C_{be} \rightarrow C_\pi$$

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HF Analysis:



$$a) f_{hi} = \frac{1}{2\pi [R_{si} \parallel R_{th} \parallel r_\pi] (C_\pi + C_m)}$$

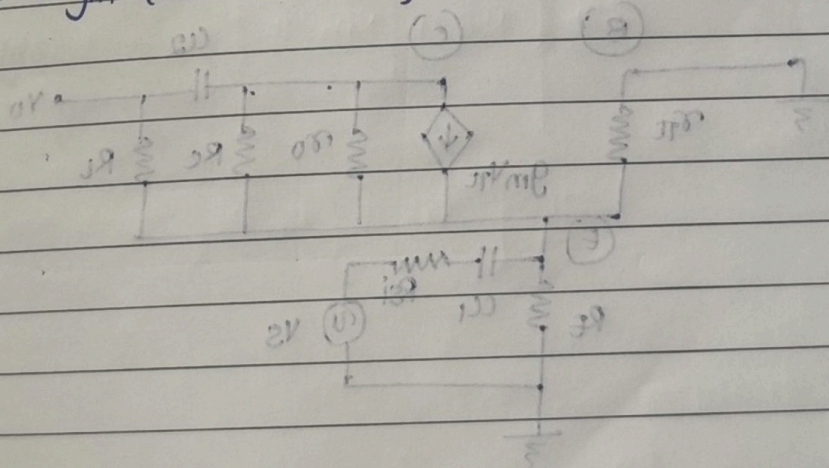
here,

$$C_m = C_u [1 + |A_{mid}|]$$

$$b) f_{ho} = \frac{1}{2\pi [R_c \parallel R_L \parallel r_o] C_u}$$

MF Analysis:

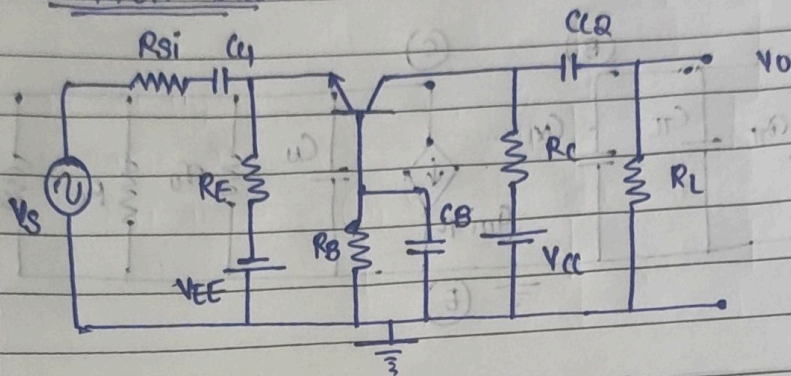
$$|A_{mid}| = -g_m (R_c \parallel R_L \parallel r_o)$$



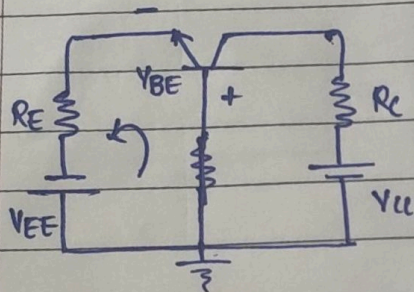
$$V_o = \frac{1}{(1 + j\omega R C)} \cdot \frac{1}{(1 + j\omega R C)}$$

$$V_o = \frac{1}{(1 + j\omega R C)} \cdot \frac{1}{(1 + j\omega R C)}$$

→ Common Base:



• Dc Analysis:



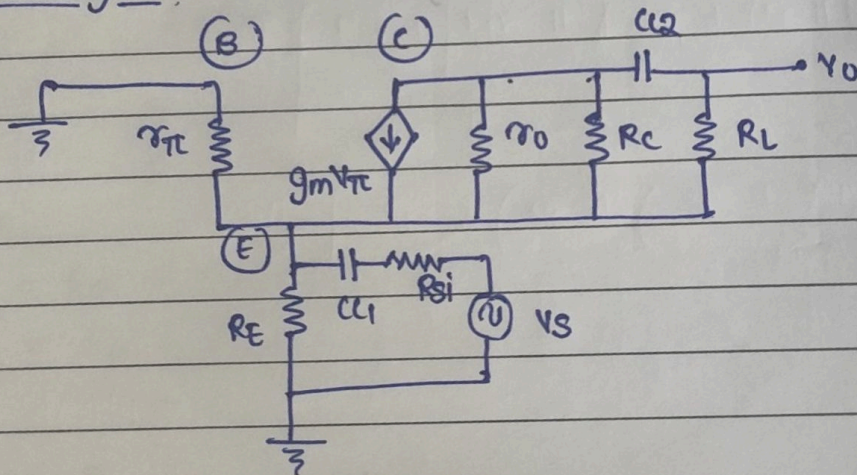
KVL on shown loop:

$$\therefore -V_{BE} - I_E R_E + V_{EE} = 0$$

$$\therefore I_E = \frac{V_{EE} - V_{BE}}{R_E}$$

thus, $I_B = \frac{I_E}{(\beta+1)}$, $I_C = \beta I_B$

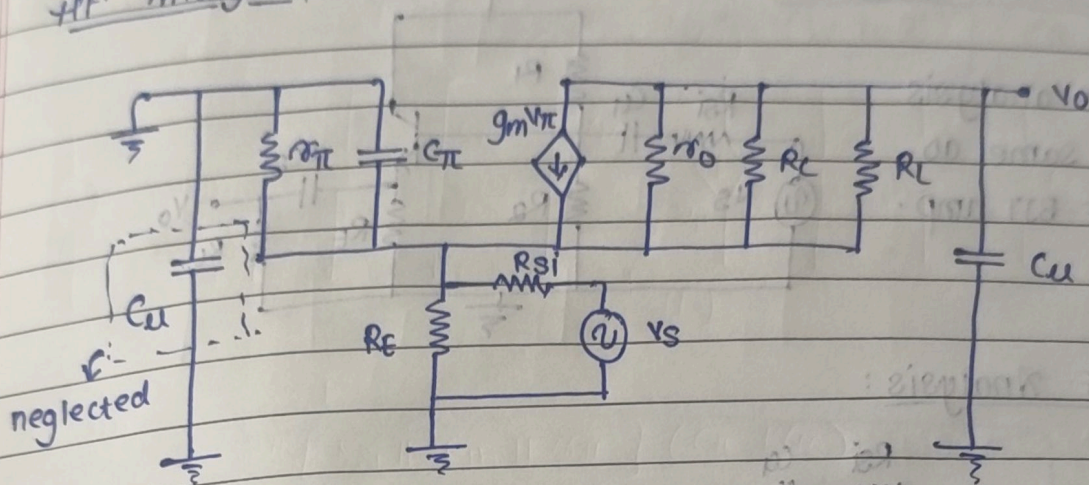
• IF Analysis:



$$a) f_{u1} = \frac{1}{2\pi \left[R_{Si} + \left(R_E \parallel \frac{r_{\pi}}{(1+\beta)} \right) \right] C_1}$$

$$b) f_{u2} = \frac{1}{2\pi \left[R_L + (R_C \parallel r_o) \right] C_2}$$

HF Analysis :



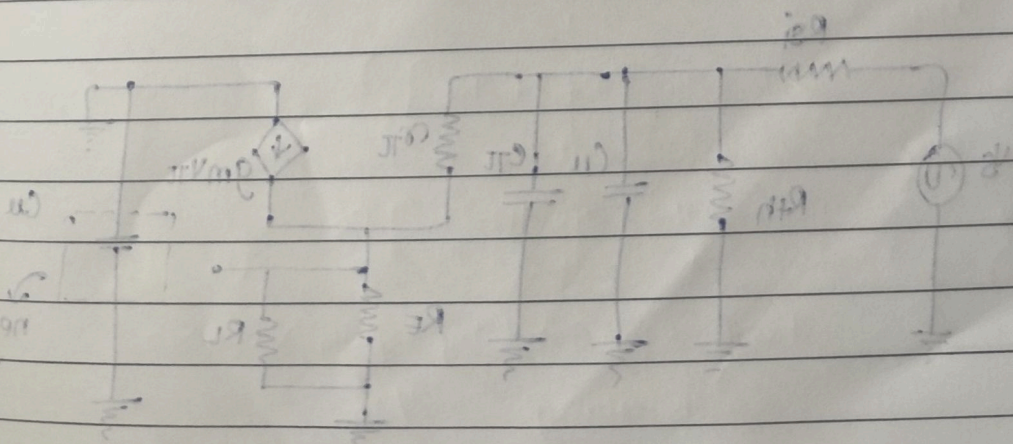
$$a) f_{hi} = \frac{1}{2\pi \left[R_{si} \parallel R_E \parallel \frac{r_{\pi}}{(1+\beta)} \right] C_{\pi}}$$

$$b) f_{ho} = \frac{1}{2\pi \left[r_o \parallel R_C \parallel R_L \right] C_u}$$

MF Analysis :

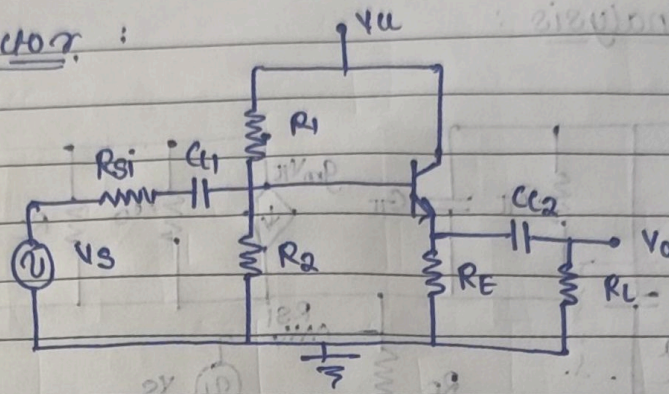
$$A_{v \text{ mid}} = g_m (r_o \parallel R_C \parallel R_L)$$

$$\left[\left(\frac{r_o \parallel R_C \parallel R_L + j\omega L}{(1+\beta)} \right) + j\omega L \right] \pi \beta$$

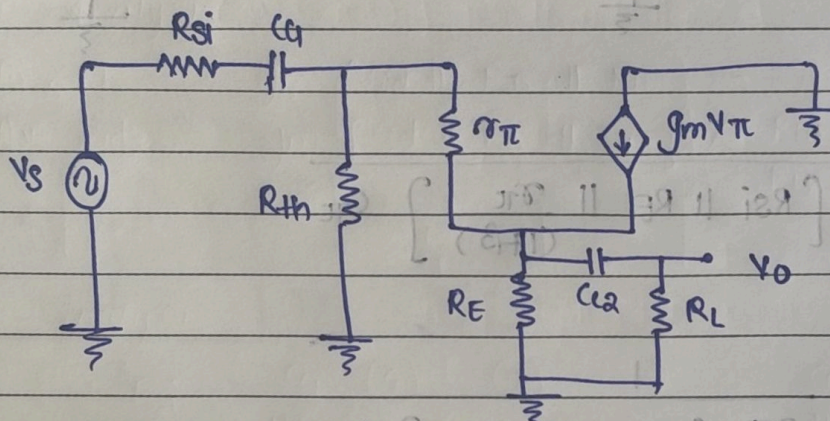


→ Common Collector :

- DC analysis is same as CE-BJT amp.



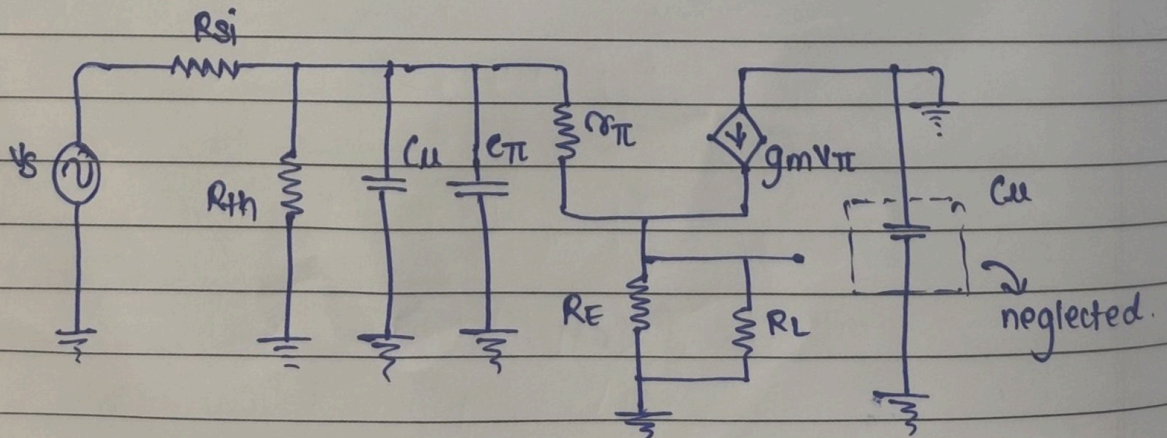
- LF Analysis :



a) $f_{u1} = \frac{1}{2\pi \left[R_{si} + \left(R_{th} \parallel \left(r_{\pi} + (R_E \parallel R_L)(1+\beta) \right) \right) \right] C_1}$

b) $f_{u2} = \frac{1}{2\pi \left[R_L + \left(R_E \parallel \left(\frac{r_{\pi} + R_{th} \parallel R_{si}}{1+\beta} \right) \right) \right] C_2}$

- HF Analysis :



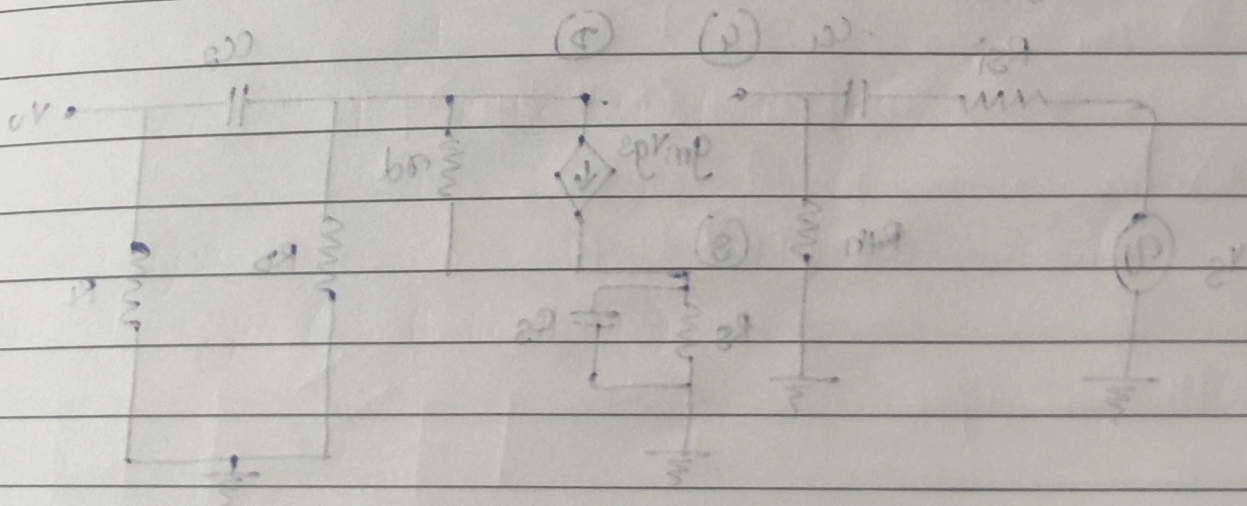
$$a) f_{HI} = \frac{1}{2\pi [R_{Si} \parallel R_{in} \parallel r_{\pi} (1 + g_m R')] \left[C_u + \frac{C_{\pi}}{1 + g_m R'} \right]}$$

MF Analysis:

$$A_v = \frac{(1 + \beta) (r_o \parallel R_E)}{r_{\pi} + (1 + \beta) (r_o \parallel R_E)} \approx 1$$

Gain: CE = CB (only phase opp.), CC = 1

BW: CE < CB < CC



$$10 \left[20 + j\omega \right] \pi B$$

$$20 \left[20 + j\omega \right] \pi B$$

$$20 \left[\frac{1}{j\omega} \parallel 20 \right] \pi B$$

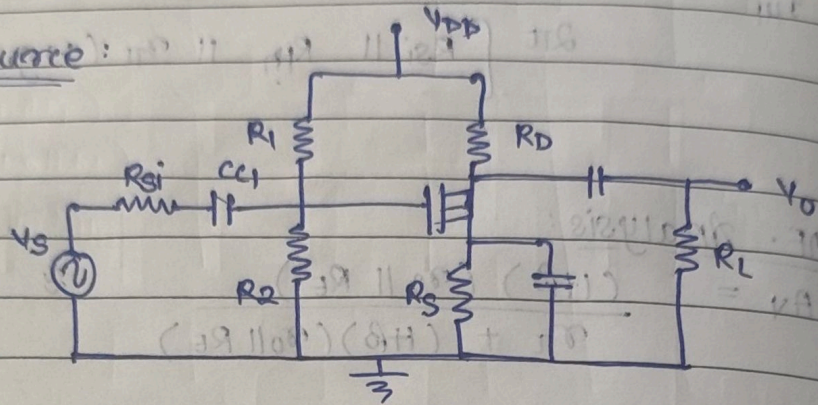
$$\begin{aligned} g_m &= 2K_n [V_{GS} - V_{TN}] \\ I_D &= K_n [V_{GS} - V_{TN}]^2 \end{aligned} \quad \begin{aligned} K_n &= \frac{K_n'}{2} \cdot \frac{W}{L} \\ r_d &= \frac{V_A}{I_D} \end{aligned}$$

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MOSFET Amplifiers

→ Common Source:

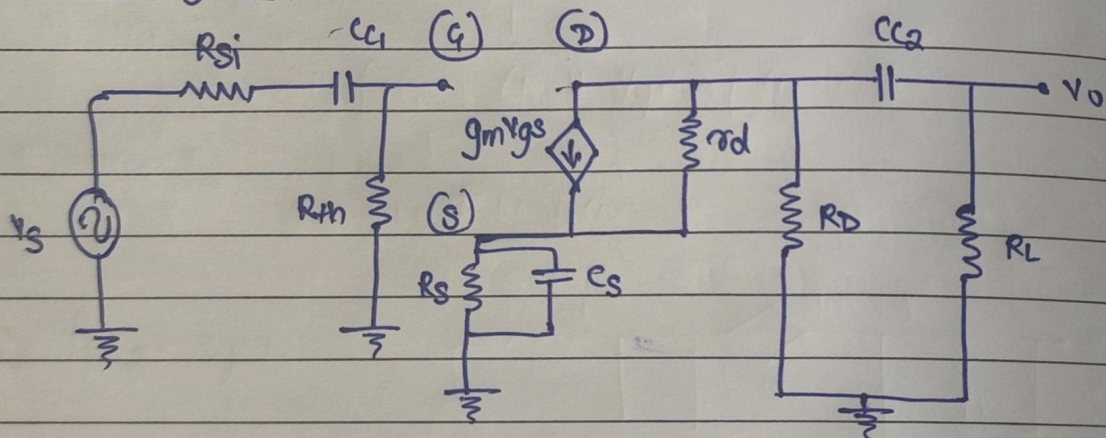


• MF Analysis:

$$A_{v \text{ mid}} = -g_m [r_d \parallel R_D \parallel R_L]$$

$$A_{v_s \text{ mid}} = -g_m [r_d \parallel R_D \parallel R_L] \left(\frac{R_{th}}{R_{th} + R_{si}} \right)$$

• LF Analysis:



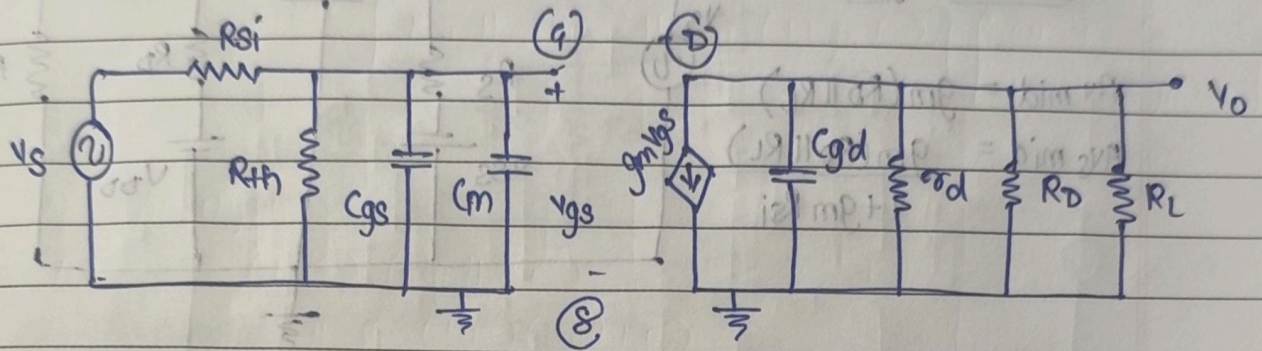
$$a) f_{u1} = \frac{1}{2\pi [R_{si} + R_{th}] C_{c1}}$$

$$b) f_{u2} = \frac{1}{2\pi [R_D + R_L] C_{c2}}$$

$$c) f_{us} = \frac{1}{2\pi [R_s \parallel \frac{1}{g_m}] C_s}$$

here, R offered by $(g_m v_{gs})$ source
is always $= 1/g_m$

• HF. Analysis:



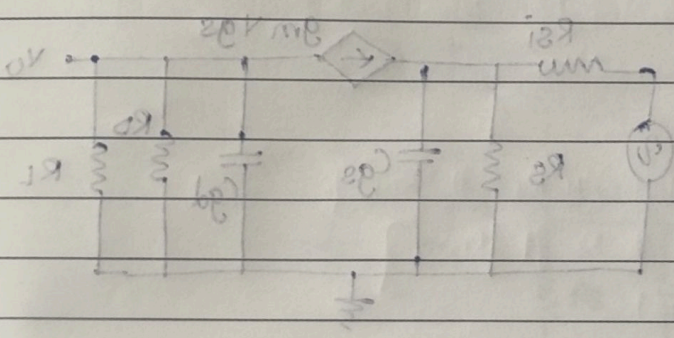
a) $f_{Hi} = \frac{1}{2\pi [R_{Si} \parallel R_{th}] [C_{gs} + C_m]}$

$C_m = C_{gd} [1 + |A_{v \text{ mid}}|]$

b) $f_{Ho} = \frac{1}{2\pi [r_d \parallel R_D \parallel R_L] C_{gd}}$

$$= \frac{1}{2\pi [(100V \parallel 120) + j20] \pi 10^{-12}}$$

$$= \frac{1}{2\pi [10 + j20] \pi 10^{-12}}$$



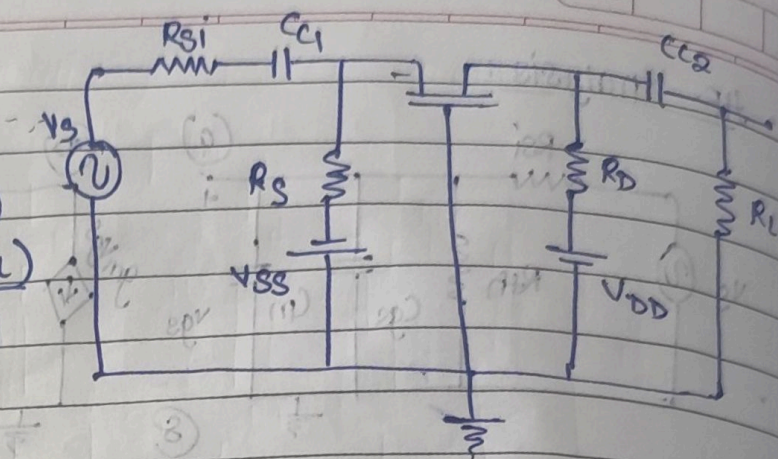
$$= \frac{1}{2\pi [100V \parallel 120 \parallel 100] \pi 10^{-12}}$$

$$= \frac{1}{2\pi [10 \parallel 10] \pi 10^{-12}}$$

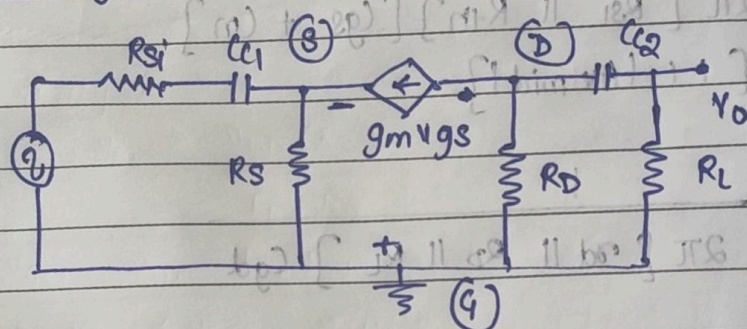
→ Common Gate :

$$A_{v \text{ mid}} = g_m (R_D \parallel R_L)$$

$$A_{v_s \text{ mid}} = \frac{g_m (R_D \parallel R_L)}{1 + g_m R_{Si}}$$



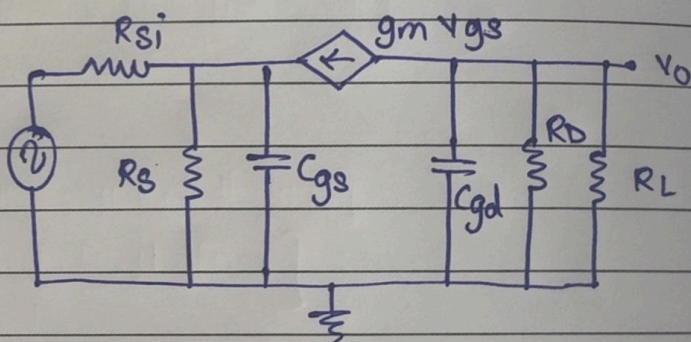
• LF Analysis :



$$a) f_{L_{C1}} = \frac{1}{2\pi [R_{Si} + (R_S \parallel 1/g_m)] C_1}$$

$$b) f_{L_{C2}} = \frac{1}{2\pi [R_D \parallel R_L] C_2}$$

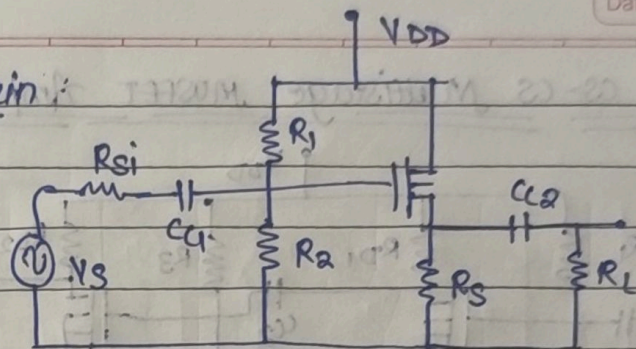
• HF analysis :



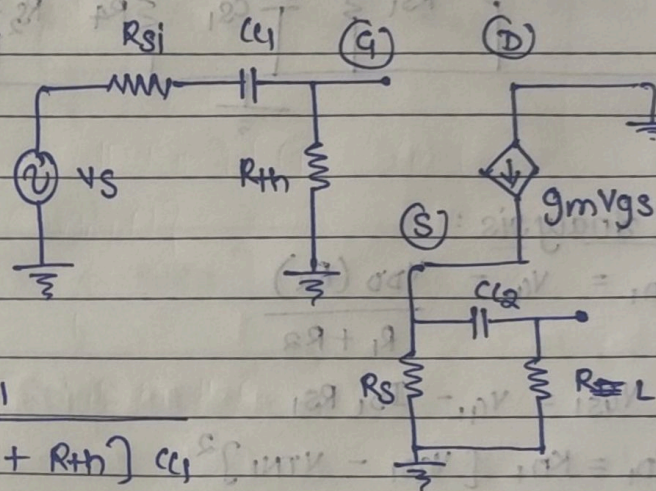
$$a) f_{Hi} = \frac{1}{2\pi [R_{Si} \parallel R_S \parallel 1/g_m] C_{gs}}$$

$$b) f_{Ho} = \frac{1}{2\pi [R_D \parallel R_L] C_{gd}}$$

→ Common Drain:



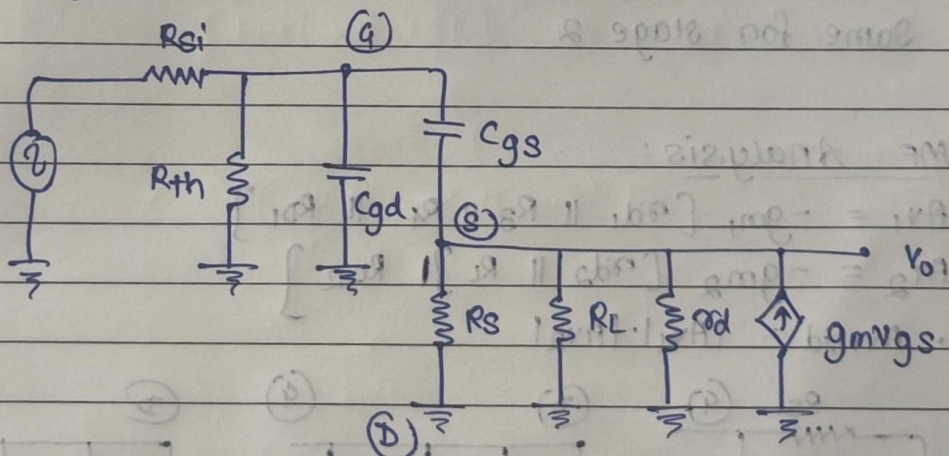
• LF Analysis:



$$a) f_{Lc1} = \frac{1}{2\pi [R_{si} + R_{th}] C_{c1}}$$

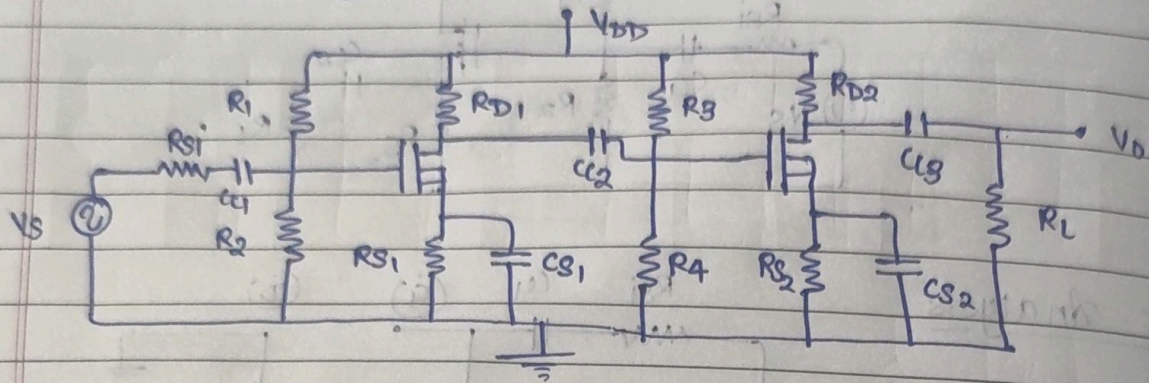
$$b) f_{Lc2} = \frac{1}{2\pi [R_L + (R_S \parallel r_d)] C_{c2}}$$

• HF Analysis:



$$a) f_{Hi} = \frac{1}{2\pi [R_{si} \parallel R_{th}] \left[C_{gd} + \frac{C_{gs}}{1 + g_m (R_S \parallel r_d \parallel R_L)} \right]}$$

→ Cascade CS-CS Multistage MOSFET Amplifier



• DC Analysis:

$$\rightarrow V_{th1} = V_{G1} = \frac{V_{DD}(R_2)}{R_1 + R_2}$$

$$\therefore V_{GS1} = V_{G1} - I_{D1} R_{S1}$$

$$I_{D1} = K_{n1} [V_{GS1} - V_{TN1}]^2$$

$$V_{DS1} = V_{DD} - I_{D1} (R_{D1} + R_{S1})$$

$$g_{m1} = 2K_{n1} [V_{GS1} - V_{TN1}]$$

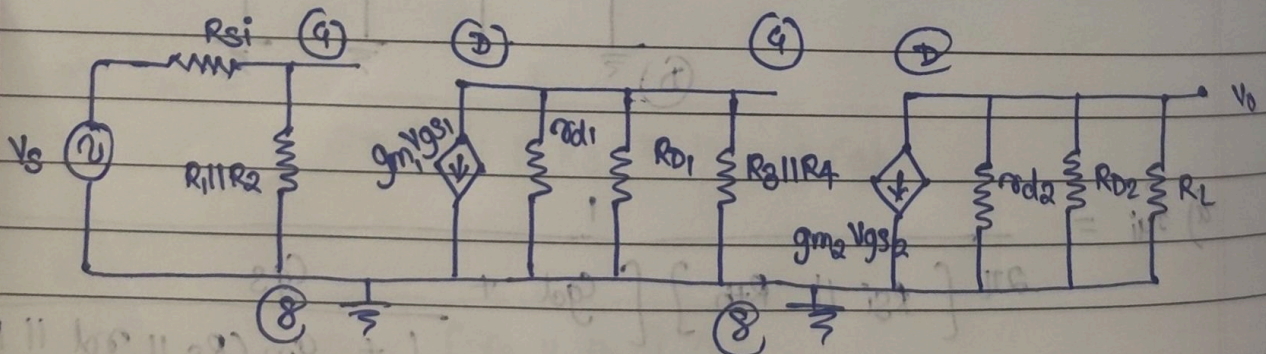
→ Same for stage 2

• MF Analysis:

$$\rightarrow A_{V1} = -g_{m1} [r_{d1} \parallel R_3 \parallel R_4 \parallel R_{D1}]$$

$$A_{V2} = -g_{m2} [r_{d2} \parallel R_L \parallel R_{D2}]$$

$$\therefore A_{Vmid} = |A_{V1}| \cdot |A_{V2}|$$

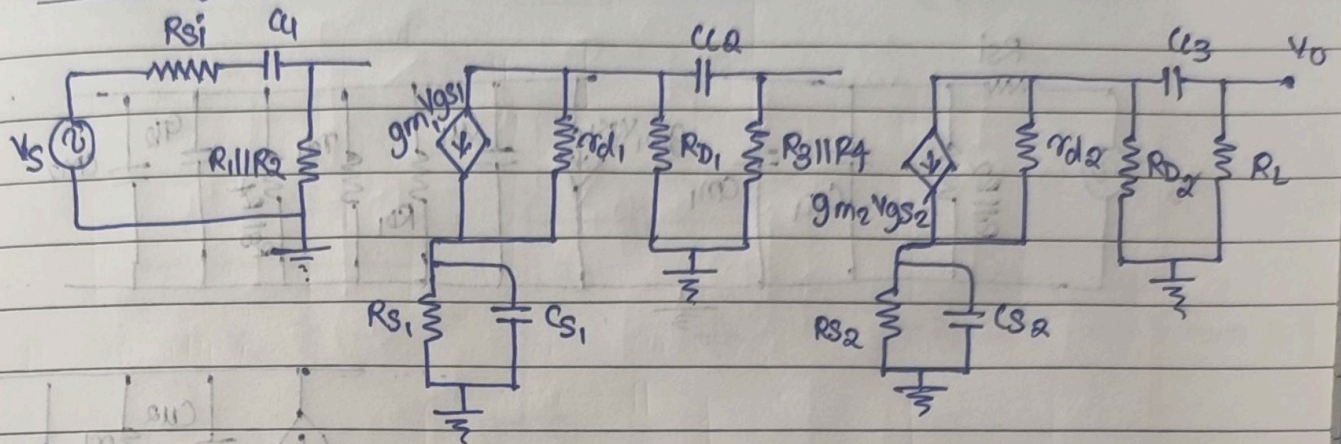


$$\rightarrow Z_{i1} = R_{Si} + (R_1 \parallel R_2)$$

$$Z_{o1} = Z_{i2} = r_{d1} \parallel R_3 \parallel R_4 \parallel R_{D1}$$

$$Z_{o2} = r_{d2} \parallel R_{D2} \parallel R_L$$

LF Analysis :



$$a) f_{u1} = \frac{1}{2\pi [R_{si} + R_1 || R_2] C_1}$$

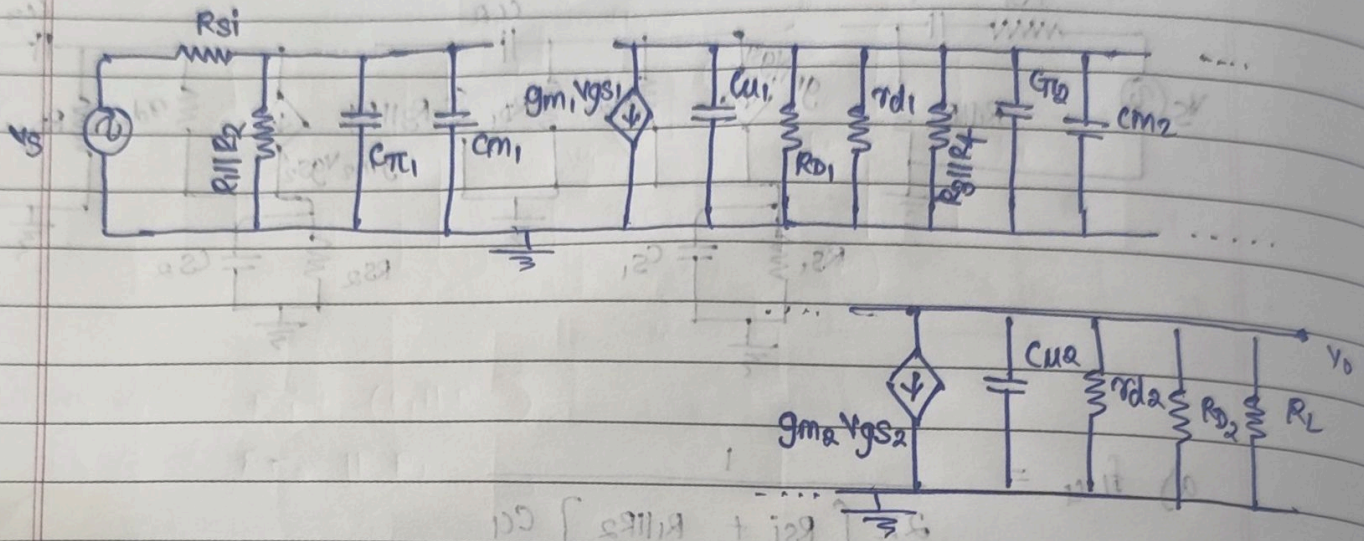
$$b) f_{u2} = \frac{1}{2\pi [R_{D1} + R_3 || R_4] C_2}$$

$$c) f_{u3} = \frac{1}{2\pi [R_{D2} + R_L] C_3}$$

$$d) f_{uS1} = \frac{1}{2\pi [R_{S1} || \frac{1}{g_{m1}}] C_{S1}}$$

$$e) f_{uS2} = \frac{1}{2\pi [R_{S2} || \frac{1}{g_{m2}}] C_{S2}}$$

HF Analysis:



$$a) f_{hi1} = \frac{1}{2\pi [R_{si} \parallel R_1 \parallel R_2] (C_{\pi 1} + C_{m1})}$$

$$\text{here, } C_{m1} = C_{u1} [1 + |A_{v1}|_{mid}]$$

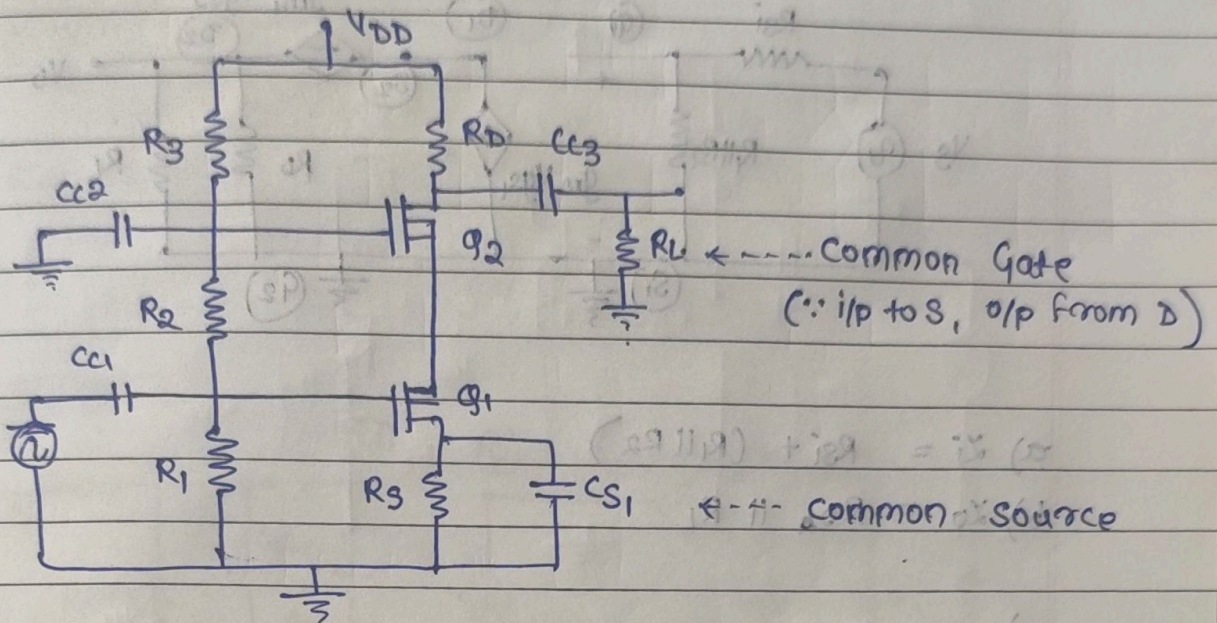
$$b) f_{HO1} = f_{hi2}$$

$$= \frac{1}{2\pi [R_{D1} \parallel r_{d1} \parallel R_3 \parallel R_4] (C_{u1} + C_{\pi 2} + C_{m2})}$$

$$\text{here, } C_{m2} = C_{u2} [1 + |A_{v2}|_{mid}]$$

$$c) f_{HO2} = \frac{1}{2\pi (r_{d2} \parallel R_{D2} \parallel R_L) (C_{u2})}$$

→ Cascode Amplifier: (CS-CG) MOSFET



- DC Analysis: Both transistors will be identical (given in question)
 $\therefore K_{n1} = K_{n2} = V_{TN1} = V_{TN2}$
 also, $I_{D1} = I_{D2}$ (also given in question)

$$\therefore V_{GS1} = V_{GS2}$$

$$\rightarrow V_{G1} = \frac{R_1 (V_{DD})}{R_1 + R_2 + R_3}, \quad V_{G2} = \frac{(R_1 + R_2) (V_{DD})}{R_1 + R_2 + R_3}$$

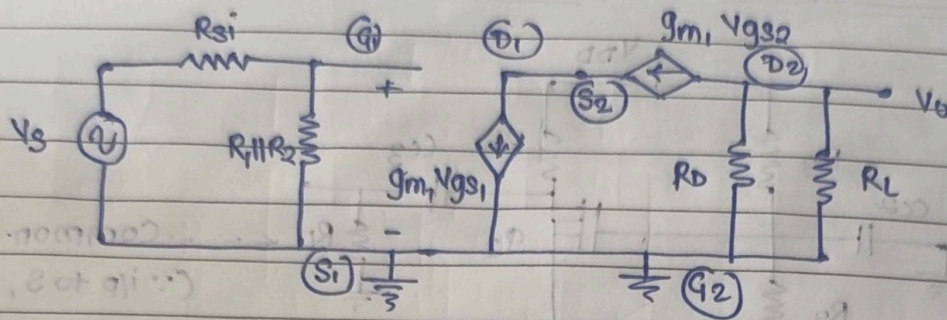
$$\rightarrow V_{D2} = V_{DD} - I_{D2} \cdot R_D$$

$$V_{S1} = I_{D1} \cdot R_S$$

$$\text{then, } V_{S2} = V_{D1}$$

$$\rightarrow g_m = 2K_n [V_{GS} - V_{TN}]$$

• MF analysis



a) $Z_i = R_{Si} + (R_1 || R_2)$

b) $Z_o = R_D || R_L$

c) For A_v :

$$A_{v1} = \frac{V_o}{V_i} = \frac{-g_{m1} V_{gs1}}{V_{gs1}} = -1$$

$$A_{v2} = \frac{V_o}{V_i} = \frac{-g_{m2} V_{gs2} (R_D || R_L)}{-V_{gs2}} = g_{m2} (R_D || R_L)$$

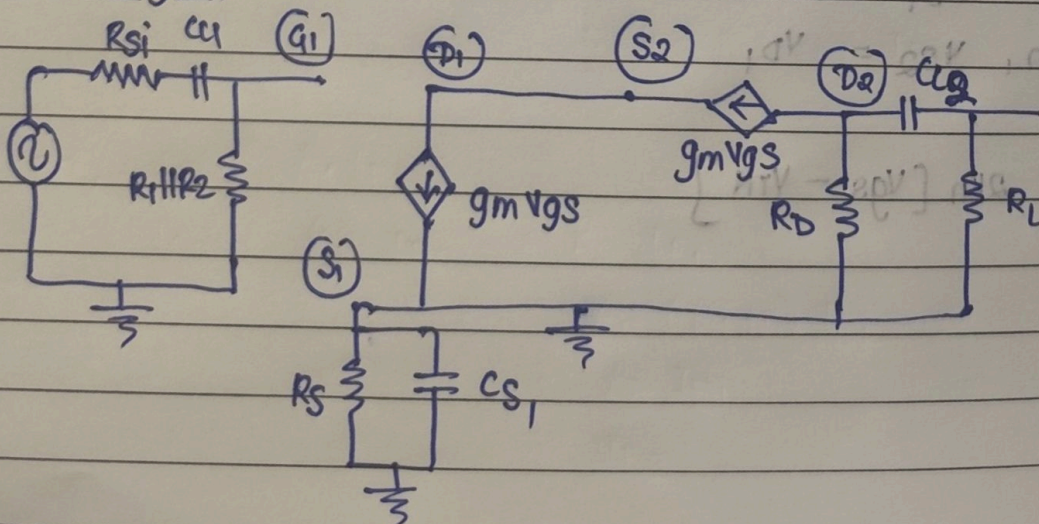
$$\therefore A_v = A_{v1} \cdot A_{v2}$$

$$= -g_{m2} (R_D || R_L)$$

here, $g_{m1} = g_{m2} = g_m$

$$\therefore A_v = -g_m (R_D || R_L)$$

• LF analysis:

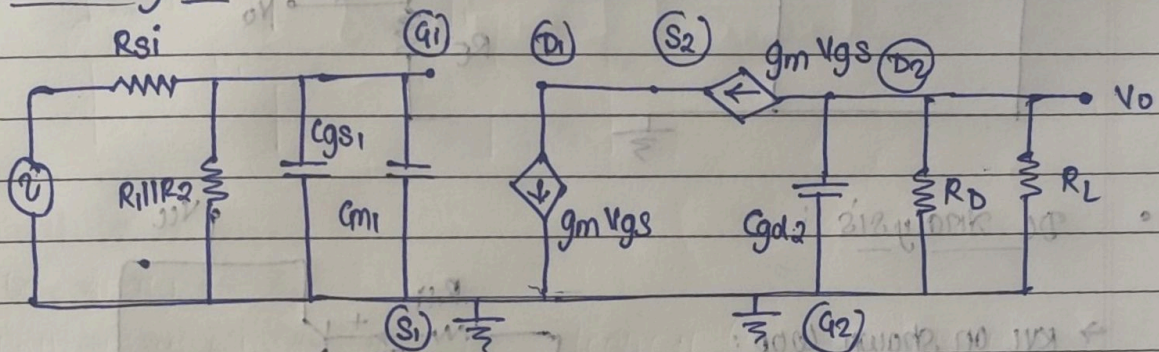


$$a) f_{LH1} = \frac{1}{2\pi (R_{Si} + R_1 \parallel R_2) C_1}$$

$$b) f_{LH2} = \frac{1}{2\pi (R_D + R_L) C_2}$$

$$c) f_{LCS1} = \frac{1}{2\pi \left[R_S \parallel \frac{1}{g_m} \right] C_1}$$

• HF analysis:



$$a) f_{H1} = \frac{1}{2\pi [R_{Si} \parallel R_1 \parallel R_2] (C_{gs1} + C_{mi})}$$

$$\text{here, } C_{mi} = C_{gd1} [1 + g_{m2} V_{GS2} (R_D \parallel R_L)]$$

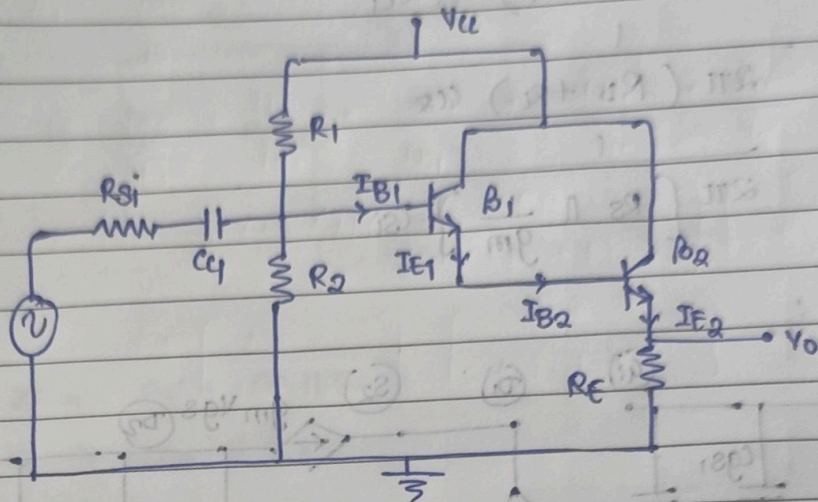
$$b) f_{H2} = \frac{1}{2\pi [R_D \parallel R_L] C_{gd2}}$$

$$\beta_D = \beta_1 \cdot \beta_2$$

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→ Darlington Amplifier (cc-cc pair):



• DC Analysis:

→ KVL on shown loop:

$$\therefore V_{th} - I_{B1} R_{th} - V_{BE1} - V_{BE2} - I_{E2} R_E = 0$$

$$\therefore I_{B1} = \frac{V_{th} - V_{BE1} - V_{BE2}}{R_{th} + \beta_D R_E}$$

$$\therefore I_{E1} = (\beta_1 + 1) I_{B1}$$

• → KVL on o/p loop:

$$\therefore V_{cc} - V_{CE2} - I_{E2} R_E = 0$$

$$\therefore V_{CE2} = V_{cc} - I_{E2} R_E$$

$$I_{E2} = (1 + \beta_2) I_{B2}$$

$$= (1 + \beta_2) I_{E1}$$

$$= (1 + \beta_2) (1 + \beta_1) I_{B1}$$

$$\therefore I_{E2} = \beta_D I_{B1}$$

$$\text{Then, } I_{C1} = \beta_1 I_{B1}$$

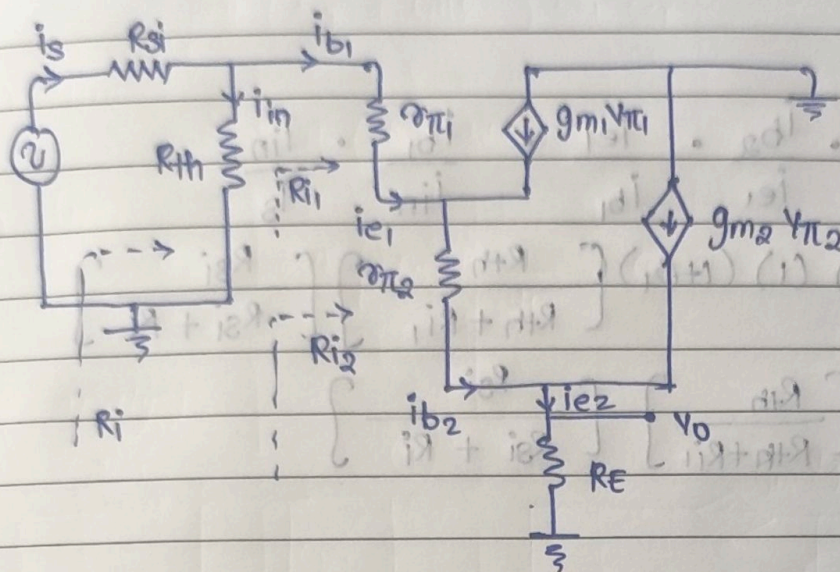
$$I_{C2} = \beta_2 I_{B2} = \beta_2 I_{E1}$$

$$= \beta_2 (\beta_1 + 1) I_{B1}$$

$$\approx \beta_D I_{B1}$$

$$\approx I_{E2}$$

MF Analysis:



a) Total i/p R:

$$R_{in} = R_{si} + R_i$$

$$= R_{si} + (R_{th} \parallel R_{i1})$$

here,

$$R_{i1} = r_{\pi 1} + (1 + \beta_1) \left[r_{\pi 2} + (1 + \beta_2) R_E \right]$$

(RRR from $r_{\pi 1}$ to R_E) (RRR from $r_{\pi 2}$ to R_E)

$$\rightarrow \therefore R_{in} = R_{si} + \left[R_{th} \parallel \left(r_{\pi 1} + (1 + \beta_1) (r_{\pi 2} + (1 + \beta_2) R_E) \right) \right]$$

b) Total o/p R:

$$R_o = R_E \parallel R_{o1}$$

$$= R_E \parallel \left[\frac{R_{o2} + r_{\pi 2}}{1 + \beta_2} \right]$$

here,

$$R_{o2} = \frac{R_{si} \parallel R_{th} + r_{\pi 1}}{1 + \beta_1}$$

$$\rightarrow \therefore R_o = R_E \parallel \left[\frac{\left(\frac{R_{si} \parallel R_{th} + r_{\pi 1}}{1 + \beta_1} \right) + r_{\pi 2}}{1 + \beta_2} \right]$$

c) Current gain:

$$A_i = \frac{i_{e2}}{i_s}$$

$$= \frac{i_{e2}}{i_{b1}} \cdot \frac{i_{b2}}{i_{e1}} \cdot \frac{i_{e1}}{i_{b1}} \cdot \frac{i_{b1}}{i_{in}} \cdot \frac{i_{in}}{i_s}$$

$$= (1+\beta_2) (1) (1+\beta_1) \left[\frac{R_{th}}{R_{th} + R_{i1}} \right] \left[\frac{R_{si}}{R_{si} + R_i} \right]$$

$$= \beta_D \left[\frac{R_{th}}{R_{th} + R_{i1}} \right] \left[\frac{R_{si}}{R_{si} + R_i} \right]$$